Synthetic Clinico-Transcriptomic Data Benchmark Report

Using differential expression, machine learning, statistical fidelity and privacy analyses to benchmark synthetic RNA-seq cohorts

Author: dbTwin, Inc.

Email: aditya@dbtwin.ai

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Executive summary

This report evaluates the quality of the synthetic cohort created from the source cohort (TCGA lung adenocarcinoma (TCGA-LUAD)) using dbTwin [1], a proprietary synthetic data engine that uses advanced ML methods to build synthetic multimodal transcriptomic data (e.g., RNA-seq counts with linked clinical metadata). This report compares the synthetic cohort against the real cohort across translational analyses: differential expression, predictive ML, gene expression statistics and privacy analyses.

Bottom line. 8 of 8 graded fidelity and utility measures passed their thresholds (Table 2). Not graded (non-anchor columns): 0 vs 1 (clinical_stk11_mut), log2FC r = 0.883, slope = 0.70, relaxed recall 69% (Table 3). Privacy is not graded: the DCR 5th-percentile ratio was 1.047 ± 0.012 (near or above 1 = no excess proximity to training records); attribute-inference results are in Table 5.

dbTwin needs two required inputs (expression.csv and metadata.csv) and, optionally, a list of anchor columns (`anchor_cols`), a user-specified set of translationally meaningful columns that the scientist is interested in. dbTwin supports 7–10 anchor columns for most datasets. Almost all ML and DE analyses on anchor columns produce high-fidelity concordance with real data; columns not explicitly declared as anchor columns produce moderate fidelity.

Table 1. Source cohort (TCGA lung adenocarcinoma (TCGA-LUAD)) used to generate the synthetic data, and the evaluation design. Files: Table 7.

Item	Value
Real samples	510
Genes	19719
Clinical variables	10
Anchor columns	`clinical_egfr_mut, clinical_age, clinical_stage_clean, clinical_kras_mut, clinical_sex`
ML target (classes)	`clinical_stage_clean` (advanced, I, II)
DE contrasts	0 vs 1 (clinical_kras_mut); 0 vs 1 (clinical_egfr_mut); 0 vs 1 (clinical_stk11_mut)
DE genes	19,719 tested; 15,775 in the top 80% by mean expression, used in DE tables and figures
Outer folds	5 (406-407 real training, 101-102 held-out real samples each)
Synthetic samples per fold	406-407

Global fidelity is assessed using maximum mean discrepancy (MMD) between real and synthetic UMAP embeddings, and by gene–gene co-expression correlation. Discovery fidelity is assessed by differential expression (DE) concordance and the train-on-synthetic, test-on-real (TSTR) paradigm, which tests whether synthetic-trained ML models generalize to held-out real subjects. SHAP concordance then compares gene-level SHAP attributions between real- and synthetic-trained models. Privacy is evaluated separately using a closest-record memorization test and two attribute-inference attacks (see Results, Table 5).

Table 2. Summary of benchmark measures with pass/fail against fixed thresholds (see FAQs). DE, UMAP, and co-expression compare the full release with the full real cohort it was generated from (in-sample). Other values are mean ± SD across outer splits on held-out real samples; SHAP uses valid splits. TRTR is the real-data reference and is not graded. DE gene recall uses the relaxed synthetic threshold (padj < 0.075, |log2FC| > 0.4). Non-anchor columns are omitted (see Bottom line). Privacy results are reported in Table 5.

Metric	Estimate	Threshold	Pass/Fail
Statistical metrics			
Gene co-expression fidelity (Sp. rho)	**0.940**	≥ 0.8	**Pass**
UMAP embedding fidelity (MMD)	**0.020**	≤ 0.1	**Pass**
DE and ML metrics			
TRTR AUC (real-data reference)	**0.573 ± 0.036**	–	–
TSTR AUC	**0.573 ± 0.044**	≥ TRTR − 0.05	**Pass**
DE log2FC Pearson r: 0 vs 1 (clinical_kras_mut)	**0.961**	≥ 0.8	**Pass**
DE gene recall (relaxed): 0 vs 1 (clinical_kras_mut)	**92%**	≥ 80%	**Pass**
DE log2FC Pearson r: 0 vs 1 (clinical_egfr_mut)	**0.970**	≥ 0.8	**Pass**
DE gene recall (relaxed): 0 vs 1 (clinical_egfr_mut)	**93%**	≥ 80%	**Pass**
SHAP Pearson r	**0.828 ± 0.056**	≥ 0.8	**Pass**

Results

Clinical fidelity, global expression structure and DE compare the full synthetic release with the full real cohort. These comparisons are in-sample: the release was generated from all real samples, which it is then compared against. Only TSTR, SHAP and the privacy tests use real samples held out from generation.

Clinical fidelity

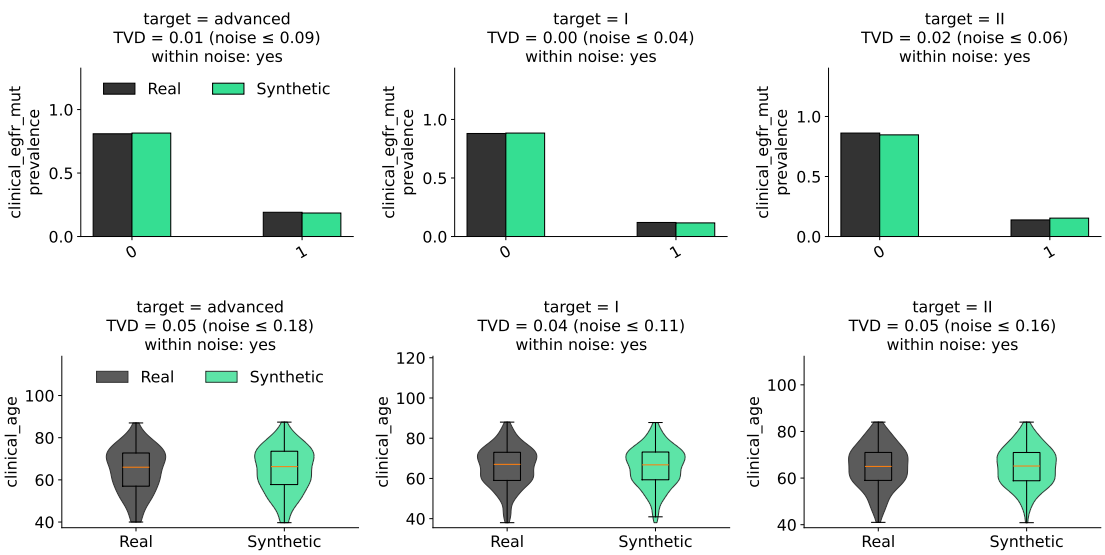


Figure 1: **Clinical fidelity of anchor columns within ML-target classes.** Full real versus full dbTwin synthetic distributions, one panel per target class: (A) `clinical_egfr_mut`; (B) `clinical_age`. Continuous columns are violins with box plots (median, IQR, whiskers to 1.5 IQR); categorical columns are level prevalences, with missing values as their own level. Panel titles give the total variation distance (TVD; 0 = identical, 1 = no overlap), using 10 equal-width bins on the real class for continuous columns. Even a perfect generator gives non-zero TVD, so each title also gives the sampling noise: the 97.5th percentile of TVD between the real class and 500 same-size bootstrap resamples of itself. "within noise: yes" means the observed TVD is no larger than that.

Global expression structure

We tested whether dbTwin preserved global statistical and embedding fidelity. A joint UMAP embedding provides a qualitative view of whether real and synthetic samples occupy comparable low-dimensional regions, while MMD summarizes separation in that projected space. Co-expression instead tests whether pairwise correlation structure among the most variable genes is retained.

Full-cohort UMAP MMD was **0.020**. The co-expression matrices had Spearman correlation **0.940**. Thus, the synthetic cohort retained the evaluated global geometry and correlation ordering to the degree quantified by these statistics. The UMAP and heat map below use the full real and synthetic cohorts. Standard per-gene RNA-seq checks (mean, variance and zero fraction of every gene, and library size per sample) are in Appendix A (Figure A1).

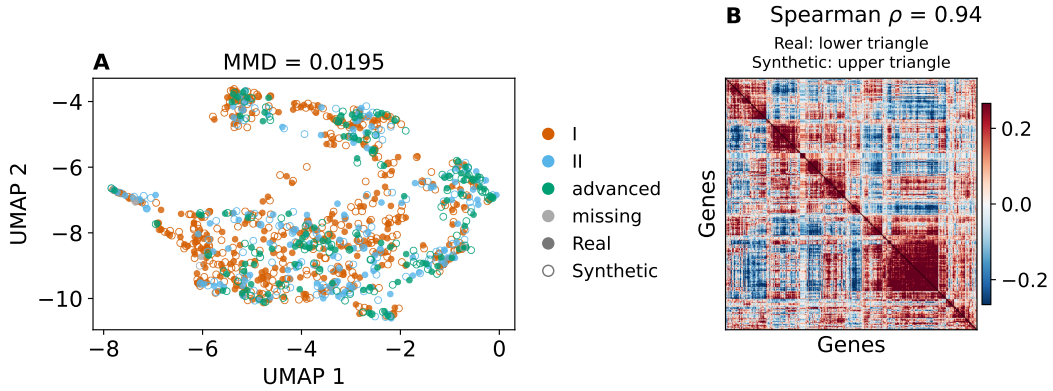


Figure 2: **Global expression structure of the full cohorts.** (A) Joint UMAP of real (dark grey) and dbTwin synthetic (green) samples, using log2(CPM + 1) (counts per million from each sample's total counts) of genes above the 40th percentile of real-cohort variance, reduced to 100 principal components. Interleaved clouds mean synthetic samples occupy the same regions as real ones; the title gives the RBF-kernel MMD between them (0 = indistinguishable; at most 0.1 passes). (B) Pearson correlations of log2(CPM + 1) for the 500 most variable real-cohort genes: real below the diagonal, synthetic above, genes in the same clustered order, so preserved modules appear as mirror-image blocks. The title gives the Spearman correlation of the off-diagonal entries (at least 0.8 passes).

Differential-expression fidelity

Differential-expression fidelity asks whether a researcher would recover comparable gene-level effects from the synthetic cohort. For each prespecified contrast, DESeq2 was applied independently to the full real and synthetic cohorts. Real DE genes have adjusted p-value below 0.05 and |log2FC| above 0.5. Agreement was evaluated using Pearson correlation of log2 fold-change among real DE genes. Synthetic DE genes use two thresholds:

- **Strict:** padj < 0.05 and |log2FC| > 0.5 (same as real).
- **Relaxed:** padj < 0.075 and |log2FC| > 0.4.

Table 3. Differential-expression agreement and DE gene counts in the full cohorts. log2FC is the first-named group relative to the second, so a positive value means higher expression in the first-named group (e.g. higher in 0 for "0 vs 1"). Real DE genes have adjusted p-value below 0.05 and |log2FC| above 0.5. log2FC r and slope are the Pearson correlation and least-squares slope of synthetic on real log2FC over all real DE genes; a slope below 1 means synthetic effects are shrunk toward zero. Synthetic DE genes use the strict or relaxed thresholds below. Recall counts real DE genes that are also synthetic DE genes, with the percentage of real DE genes recalled in brackets. Total counts all synthetic DE genes; Precision is the recalled count divided by Total. Per-gene results are in the CSV files listed in Table 7.

Contrast	log2FC r	log2FC slope	Real DE genes	Synthetic DE genes		
					Strict padj < 0.05 |log2FC| > 0.5	**Relaxed** padj < 0.075 |log2FC| > 0.4
0 vs 1 (clinical_kras_mut) positive = higher in 0	**0.961**	**0.94**	1121	Recall	**951 (85%)**	**1035 (92%)**
				Total	1125	1632
				Precision	**85%**	**63%**
0 vs 1 (clinical_egfr_mut) positive = higher in 0	**0.970**	**0.92**	1369	Recall	**1186 (87%)**	**1267 (93%)**
				Total	1397	1990
				Precision	**85%**	**64%**
0 vs 1 (clinical_stk11_mut) non-anchor feature positive = higher in 0	**0.883**	**0.70**	2586	Recall	**1519 (59%)**	**1786 (69%)**
				Total	1873	2471
				Precision	**81%**	**72%**

The reported correlations and slopes use all real DE genes, regardless of synthetic significance, so they test whether real effect sizes are reproduced: r measures agreement in ordering, while the slope shows whether effects keep their real size. When counting synthetic DE genes, the relaxed threshold (padj < 0.075 and |log2FC| > 0.4) is recommended to account for weaker significance and reduced statistical power, particularly in minority classes. The panels below show all real DE genes in the full cohorts; each subtitle reports their Pearson correlation and slope. Per-gene real and synthetic log2FC, adjusted p-values and DE calls for every tested gene are saved as one CSV per contrast (Table 7).

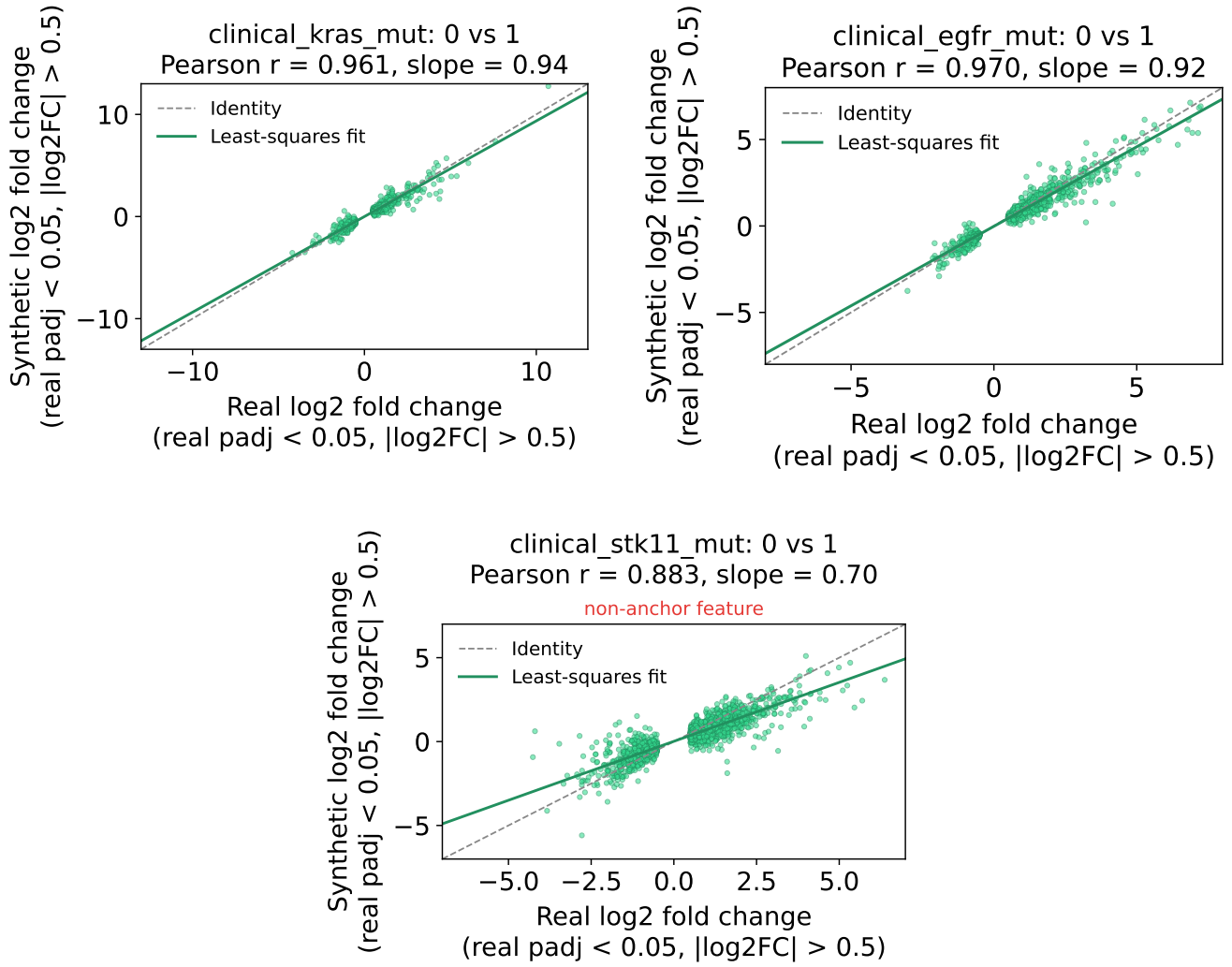
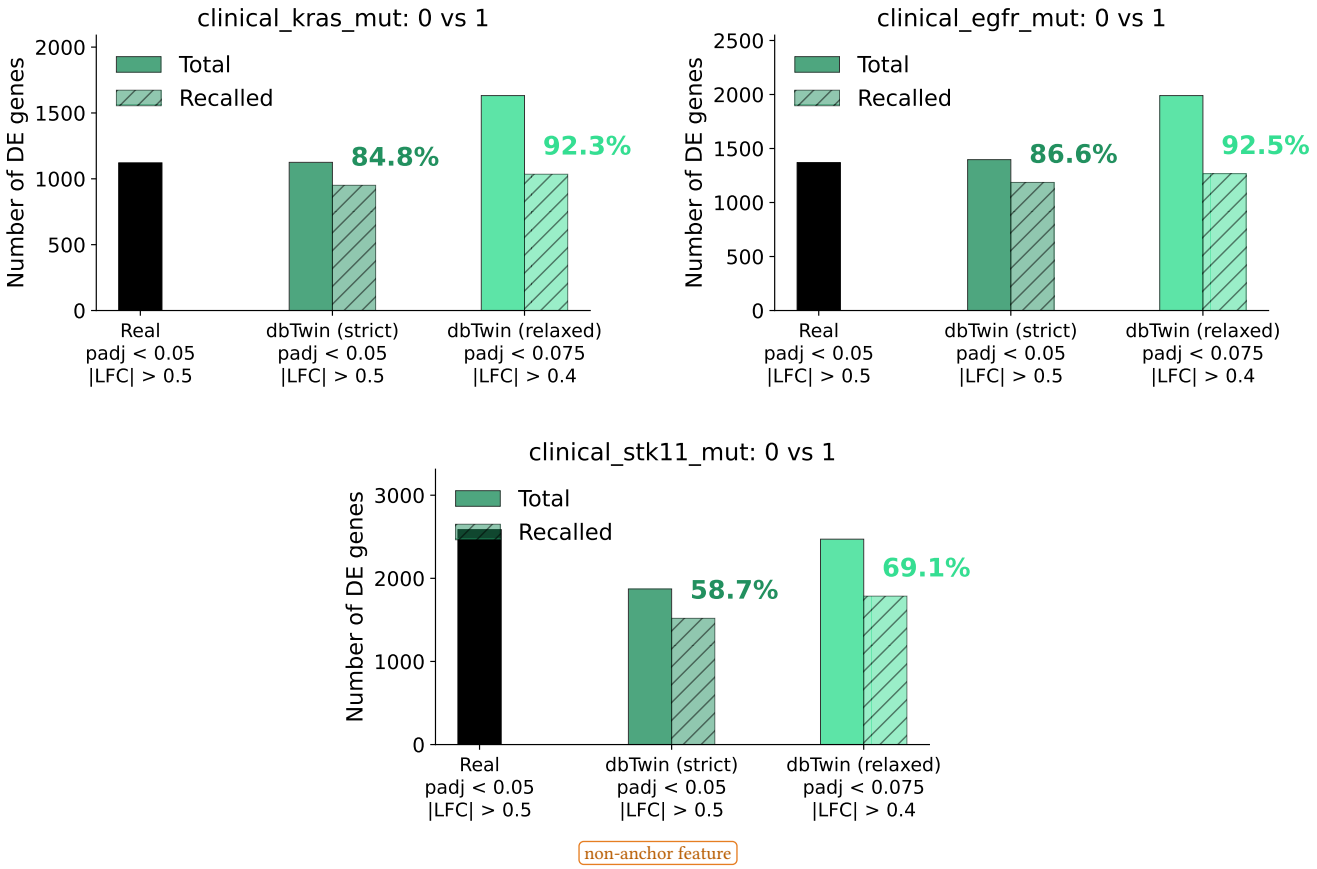


Figure 3: **Effect-size concordance of differential expression in the full cohorts.** One panel per contrast, in the order of Table 3. DESeq2 was fitted separately to the full real and synthetic cohorts on all 19,719 genes the 15,775 genes in the top 80% by real mean CPM are shown. Each point is a real DE gene (real padj < 0.05 and |log2FC| > 0.5) at its real (x) and synthetic (y) log2 fold change, first-named group relative to the second (positive = higher in the first-named group, e.g. higher in 0 for "0 vs 1"). Dashed line, identity: points on it mean real effects are reproduced in direction and size. Solid line, least-squares fit; its slope (subtitle) below 1 means synthetic effects are shrunk toward zero. Subtitles give Pearson r (at least 0.8 passes) and the slope. Non-anchor feature marks a column not supplied to dbTwin as an anchor.



non-anchor feature

Figure 4: **Recall of real DE genes in the full cohorts.** One panel per contrast, in the order of Table 3. Black: real DE genes (padj < 0.05 and |log2FC| > 0.5). Green pairs, at strict (padj < 0.05, |log2FC| > 0.5; darker) and relaxed (padj < 0.075, |log2FC| > 0.4; lighter) synthetic thresholds: all synthetic DE genes (solid) and those that are also real DE genes (hatched, recalled). Percentages give recall, the share of real DE genes recalled; precision is in Table 3. A solid bar well above its hatched bar indicates synthetic-only discoveries. The relaxed threshold is recommended for small subgroups; recall of at least 80% under this threshold passes.

ML fidelity: train-on-synthetic, test-on-real (TSTR) AUC and SHAP concordance

Predictive utility was evaluated using train-on-synthetic, test-on-real (TSTR) analysis. Elastic-net logistic regression classifiers trained on each synthetic cohort were evaluated on the corresponding held-out real fold. Train-on-real, test-on-real (TRTR) models used the same held-out subjects and provide the real-data reference. Mean ROC-AUC was **0.573 ± 0.036** for TRTR and **0.573 ± 0.044** for TSTR (mean ± SD over outer splits; each split is shown in the figure below). The comparison therefore tests out-of-sample analytical utility rather than the ability to fit synthetic records themselves.

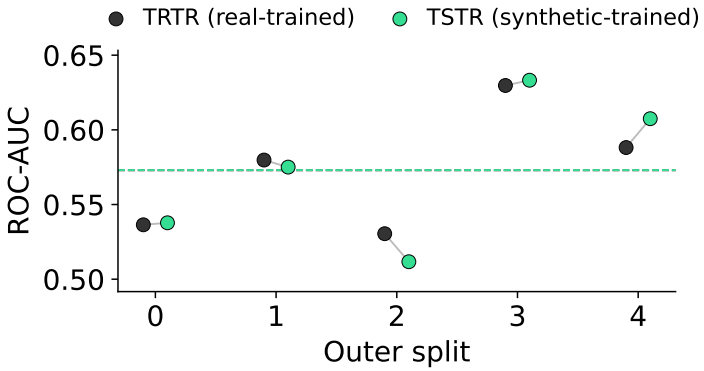


Figure 5: **TRTR and TSTR ROC-AUC in each outer split.** Each pair of dots is one split: the real-trained (dark grey) and synthetic-trained (green) elastic-net classifier for `clinical_stage_clean`, both scored on the same held-out real fold. Dashed lines give the means reported above. TSTR dots level with their TRTR partners mean synthetic-trained models generalize to new real patients as well as real-trained ones.

Matching AUC is not sufficient for transcriptomic discovery: two classifiers can discriminate the same classes while relying on different genes. We therefore compared mean absolute SHAP values from real-trained and synthetic-trained classifiers, with both explanations evaluated on held-out real samples. Correlations were computed for the 50 genes with the largest real-trained mean absolute SHAP values. Class-specific results are reported below because averaging across classes can obscure minority-class failures. Synthetic SHAP signal was absent in 0 of 5 outer folds despite variable real SHAP signal.

Table 4. Class-specific concordance of real-trained and synthetic-trained SHAP gene attributions; correlations are mean ± SD over valid splits. Top-10 genes shared: how many of the ten genes with the highest real-trained mean |SHAP| are also among the synthetic-trained top ten, averaged over all splits.

ML target class	SHAP Pearson r	SHAP Spearman rho	Top-10 genes shared	Splits
I	**0.833 ± 0.063**	0.769 ± 0.049	6.8/10	5
II	**0.808 ± 0.062**	0.752 ± 0.066	7.2/10	5
advanced	**0.843 ± 0.042**	0.746 ± 0.026	7.6/10	5

These correlations test concordance of the evaluated classifier's feature attributions, not causal gene effects. They are nevertheless a stringent check that synthetic-trained models recover the same predictive signal when applied to unseen real subjects. The plots show the nine leading real-trained genes for a valid split in which at least one plotted gene has non-zero real and synthetic SHAP values for every class; annotations summarize correlations as mean ± SD across valid outer splits.

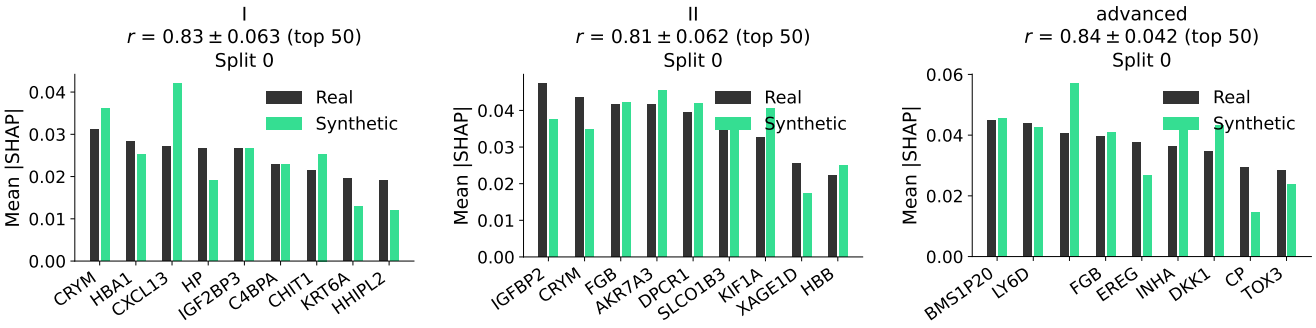


Figure 6: **Gene-level drivers of real- and synthetic-trained classifiers, representative split 0.** Elastic-net models for `clinical_stage_clean` were trained on the real training fold (dark grey) or the matching synthetic fold (green), using the same genes and regularization, and both were explained on the same held-out real subjects. One panel per class (one for binary targets). Bars show mean absolute SHAP values for the nine genes with the highest real-trained importance in this split; similar heights mean the synthetic-trained model relies on the same genes. Titles give the Pearson r over the top 50 real genes as mean ± SD across valid splits (Table 4).

Empirical privacy evaluation

Privacy was evaluated using two distinct empirical tests. DCR measures memorization-like proximity in expression space. Each synthetic sample's distance to its nearest real training sample was compared with the reference distribution of distances from each real training sample to its nearest other training sample. The ratio of the fifth percentiles of these distributions was 1.047 ± 0.012. Ratios well below 1 suggest unusually close synthetic records relative to the spacing between distinct training records; this is an empirical memorization check, not a privacy guarantee.

Attribute inference addresses a different threat model: the attacker has the synthetic data and some features of a real patient, called auxiliary columns, and tries to guess that patient's unknown features, called secrets. To separate leakage from genuine population-level biology, the same attack is repeated against held-out real patients who were not used for generation (the control). Attribute inference is a concern when the training attack success rate is substantially higher than the control success rate.

We tested two threat models. In the first, the attacker knows only the clinical quasi-identifiers (`clinical_age`, `clinical_sex`); in the second, the attacker also knows the top 100 most variable genes, ranked by variance of log2(CPM + 1). In both attacks the secrets are the remaining clinical columns and the next 100 most variable genes, and each secret column was attacked independently. The reported statistic is training minus control attack success [2], averaged across splits per secret column, then summarized as the median of the 10 highest column means for each attack. All attribute inference attacks were implemented using the Anonymeter package [3].

Table 5. Summary of privacy measures; DCR: mean ± SD across splits; attribute inference: median of the top 10 column means across splits.

Metric	Estimate	Features with mean excess > 0.2
DCR 5th percentile ratio	1.047 ± 0.012	--
100 expression genes + QI median of top 10 means	0.143	3
Clinical QI median of top 10 means	0.028	0

Feature counts in Table 5 include all secret columns whose mean training minus control success across splits is strictly above 0.2, separately for each attack. Excess alone does not show how successful an attack is, so Table 6 gives the absolute success rates behind the largest excesses.

Table 6. Secrets with the five largest mean excesses for each attack. Attack success rates are averaged over outer splits: training (attacking patients used for generation) and control (held-out patients). Excess is training minus control.

Attack	Secret	Training	Control	Excess
Clinical QI + 100 genes	`clinical_stage_clean`	0.94	0.42	**0.52**
Clinical QI + 100 genes	`clinical_kras_mut`	0.96	0.64	**0.31**
Clinical QI + 100 genes	`clinical_tp53_mut`	0.91	0.62	**0.29**
Clinical QI + 100 genes	`clinical_os_months`	0.21	0.05	**0.16**
Clinical QI + 100 genes	`clinical_egfr_mut`	0.96	0.82	**0.14**
Clinical QI only	`clinical_stage_clean`	0.48	0.37	**0.12**
Clinical QI only	`clinical_kras_mut`	0.63	0.59	**0.05**
Clinical QI only	`clinical_tp53_mut`	0.55	0.52	**0.03**
Clinical QI only	`ALPL (gene)`	0.05	0.02	**0.03**
Clinical QI only	`GLB1L3 (gene)`	0.07	0.04	**0.03**

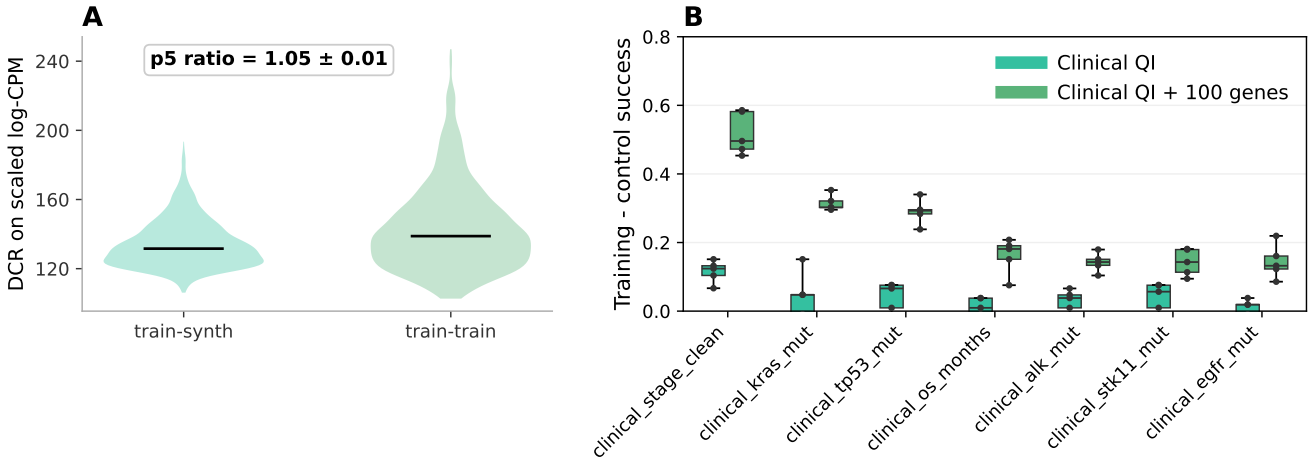


Figure 7: **Empirical privacy tests.** (A) Distance to closest record (DCR), pooled over outer splits: each synthetic record to its nearest real training record (left) versus each training record to its nearest other training record (right), on standardized log2(CPM + 1) expression. Copied or lightly perturbed records would pull the left violin toward zero; the annotation gives the 5th-percentile ratio (mean ± SD across splits), where values near or above 1 show no excess proximity. (B) Attribute inference (Anonymeter): an attacker who knows `clinical_age, clinical_sex` alone, or these plus 100 expression genes, infers a withheld clinical or gene secret from the closest synthetic record. The y-axis is success on training patients minus success on held-out patients, isolating what the synthetic data reveals about the individuals used to build it. The seven secrets with the highest mean excess are shown; boxes span splits, dots are splits, and the axis starts at 0.

The two attack estimates describe training-specific advantage under the implemented auxiliary information and warrant consideration in release decisions. Importantly, a high attribute inference score does not automatically raise re-identification risk: that also depends on the sensitivity of the attacked secret and of the patient population [2]. DCR tests close record replication, whereas attribute inference tests attack advantage; the two results are not interchangeable.

How to read and use this report (FAQs)

What do Pass and Fail mean in Table 2? Each metric is compared with a fixed threshold: co-expression rho at least 0.8; UMAP MMD at most 0.1; TSTR AUC within 0.05 of TRTR AUC; DE log2FC r at least 0.8; DE gene recall (real DE genes recalled at the relaxed synthetic threshold: padj < 0.075 and |log2FC| > 0.4) at least 80%; SHAP Pearson r at least 0.8. Non-anchor columns are exempt from the thresholds and are omitted from Table 2; their DE contrasts are listed as not graded in the Bottom line.

What does TVD mean in the clinical fidelity figure? TVD is the fraction of probability mass that differs between real and synthetic distributions (0 = identical, 1 = disjoint). Numeric columns are binned using fixed-width bins defined on the real cohort. Small classes give non-zero TVD from sampling alone, so judge each value against the sampling-noise bound in its panel title.

How do I interpret the co-expression matrix? Real gene–gene correlations are shown below the diagonal and synthetic correlations above it, in the same gene order. A Spearman rho close to 1 means the synthetic data has the same gene correlation structure as the real data.

What does the UMAP MMD mean? MMD (maximum mean discrepancy) measures the separation between real and synthetic samples in the joint UMAP embedding. An MMD of 0 means real and synthetic samples are indistinguishable in that projection; larger values indicate increasing distance.

How should I compare TSTR and TRTR? TRTR is the real-data reference. TSTR close to TRTR means models trained on synthetic data achieve similar AUC on held-out real subjects as models trained on real data.

What does the DE log2FC r show? It correlates real and synthetic effect sizes for all real DE genes (real padj < 0.05 and |log2FC| > 0.5). A value close to 1 means the synthetic cohort reproduces the same direction and ordering of effects. The slope adds magnitude: near 1, synthetic effects keep their real size; below 1, they are shrunk toward zero even when r is high.

How do I read the DE gene counts? Recall is the share of real DE genes that are also significant in the synthetic cohort (recalled count in Table 3); Total is all synthetic DE genes. A recalled count close to the real DE count means real discoveries are recovered, while Total far above it indicates synthetic-only discoveries. Precision is the recalled count divided by Total: the share of synthetic DE genes that are also real DE genes.

What does SHAP concordance show? A SHAP r close to 1 means synthetic-trained classifiers rely on the same genes, with similar relative importance, as real-trained classifiers.

How do I read the privacy results? The DCR ratio divides the fifth percentile of synthetic-to-training nearest-neighbor distances by the fifth percentile of training-to-other-training nearest-neighbor distances. Ratios near 1 indicate comparable lower-tail distances; much lower ratios suggest memorization. Attribute-inference excess success near 0 means the synthetic data gives an attacker no advantage beyond population-level knowledge.

How should I use the synthetic cohort? For exploratory analysis, hypothesis generation, workflow development, and preliminary feature prioritization, followed by confirmation on governed real data (see Limitations).

Methods

Clinical fidelity, differential expression, UMAP, and co-expression are evaluated once using the full real and synthetic cohorts. These comparisons are in-sample: the full synthetic release was generated from every real sample, and it is compared against those same samples. Predictive ML accuracy, SHAP concordance, DCR and attribute inference are evaluated using a 5-fold train-test split with synthetic datasets generated only from the training folds, so ML accuracy, SHAP and the control arm of attribute inference use real samples held out from generation. All synthetic folds are saved in `datasets/<data-folder>/dbTwin/artifacts/` and only used the training folds as delineated in the split_manifest.

All real data is stored in `datasets/<data-folder>/real_data`.

Global and clinical fidelity

Clinical anchor-column distributions were compared within ML-target classes using total variation distance (TVD) in the full real and synthetic cohorts. As a sampling-noise reference, each real class was compared with 500 bootstrap resamples of itself, each the size of the matching synthetic class; an observed TVD at or below the 97.5th percentile of these values is within sampling noise. Each column was treated as categorical or continuous once, from its values in the full real cohort, for every class and for the noise reference. Expression analyses used log2(CPM + 1), with counts per million (CPM) from each sample's total counts over all genes. For co-expression, genes were ranked by variance of these values in the full real cohort, and Pearson gene–gene correlation matrices of the top 500 genes were computed separately in real and synthetic data. Average-linkage hierarchical ordering derived from the real matrix was applied to both. Spearman correlation between corresponding upper-triangle entries provided the scalar co-expression statistic.

For UMAP, the full real and synthetic cohorts were transformed to log2(CPM + 1), so that sequencing depth does not drive the embedding, and genes at or below the 40th percentile of full-real-cohort variance were excluded. The concatenated cohorts were reduced to 100 principal components (centred, not scaled) and embedded in a single two-dimensional UMAP fit with 15 neighbors, minimum distance 0.1, Euclidean distance, and random state 42. MMD was computed with an RBF kernel on these two-dimensional coordinates. These analyses were computed once for the full cohorts.

Differential expression

All 19,719 genes were tested. DESeq2 models were then fit independently to rounded integer counts from the full real and synthetic cohorts using each contrast's specified clinical column as the design factor. As in R DESeq2, counts flagged as outliers by Cook's distance were replaced and those genes refit (levels with at least seven samples); log2 fold-changes were not shrunk. Adjusted p-values and log2 fold-changes were aligned by gene. DE was run on all subjects, but only the top 80% of genes with the highest mean CPM in the real cohort (15,775 of 19,719 tested genes) were considered in the figures and tables. Real DE genes were those with adjusted p-value below 0.05 and |log2FC| above 0.5 in the real cohort; the Pearson correlation and least-squares slope of synthetic on real log2 fold-change used all of them. log2 fold-changes are the first-named group relative to the second. Synthetic DE genes use two thresholds:

- **Strict:** padj < 0.05 and |log2FC| > 0.5.
- **Relaxed:** padj < 0.075 and |log2FC| > 0.4.

Table 3 reports total synthetic DE genes, recall of the real DE set (recalled count and percentage), and precision (recalled count divided by total). Each scatter plot shows one contrast in the full cohorts, with its Pearson r and slope in the subtitle.

Machine learning and SHAP

The classifier was elastic-net logistic regression with balanced class weights and the SAGA solver. In each outer training cohort, the number of expression features was the minimum of 270, 90% of the training sample count, and the available gene count. Counts were transformed to log2(CPM + 1), genes were ranked by training-set variance of these values, missing values were median-imputed, and features were standardized. The SAGA solver used random state 42.

Regularization was selected using only real training data. Within every outer training cohort, stratified cross-validation used up to five folds, limited by the smallest class. The l1 ratio was fixed at 0.05 and C was evaluated at 51 log-spaced values from 10^{-4} to 10^{1}. A candidate was eligible only when the active-feature fraction lay between 20% and 60% in every inner fold of every outer split. The eligible value with the highest fold-weighted mean AUC was selected, with smaller C preferred on ties, and then applied unchanged to real- and synthetic-trained models.

TRTR and TSTR classifiers were evaluated on the same held-out real subjects. For multiclass ML targets, the reported ROC-AUC is the mean of one-versus-rest class AUCs. Linear SHAP values were computed on the standardized held-out real features. Within each class and split, mean absolute SHAP values were compared between real- and synthetic-trained classifiers over the top 50 genes selected by real-trained importance using Pearson and Spearman correlation. Correlations were defined only when both selected importance vectors were finite and had positive standard deviation; undefined folds were excluded from mean and SD. Signal failures counted unique folds where any class had variable real top-50 importance but all-zero synthetic importance.

Privacy analyses

For DCR, all genes were transformed to log2(CPM + 1), missing values were set to zero, and genes were standardized using the real training mean and standard deviation. Each synthetic sample's Euclidean distance to its nearest real training sample was computed. The reference distribution consisted of each real training sample's Euclidean distance to its nearest other real training sample, excluding the sample itself. For each split, the fifth percentile of the synthetic-to-training distances was divided by the fifth percentile of the real-training reference distances. The report gives the mean and standard deviation of these ratios across splits. Held-out samples and reference-set subsampling were not used in this DCR calculation.

Attribute inference used Anonymeter's inference evaluator at 95% confidence, with one attack per held-out subject. The configured clinical quasi-identifiers were `clinical_age`, `clinical_sex`. Eligible clinical secrets were `clinical_os_months`, `clinical_os_status`, `clinical_egfr_mut`, `clinical_kras_mut`, `clinical_alk_mut`, `clinical_tp53_mut`, `clinical_stk11_mut`, `clinical_stage_clean`. Genes were ranked by log2(CPM + 1) variance in each fold's synthetic cohort. Gene secrets were the next 100 ranked genes after the 100 auxiliary genes, with no overlap, and kept their released raw counts. The expression-assisted attack used the auxiliary genes as log2(CPM + 1) standardized with real training statistics; the second attack used clinical quasi-identifiers alone. The reported excess success is the training attack rate minus the control attack rate, averaged across splits per secret column, then summarized as the median of the 10 highest column means for each attack. This is distinct from Anonymeter's normalized inference-risk estimator.

Data files and layout

Paths in Table 7 are relative to `datasets/tcga_luad/dbTwin/` in the benchmark repository. On Hugging Face (NandaAditya/rna-twin) the synthetic CSVs sit under `tcga_luad/` with the same names and `artifacts/` subfolder.

Table 7. Released and benchmark files for TCGA lung adenocarcinoma (TCGA-LUAD).

File	Contents
`syn_tcga_luad.csv`	**Full synthetic release** (510 samples), generated once from all 510 real samples. This is the cohort intended for reuse.
`artifacts/syn_tcga_luad_fold_<1-5>.csv`	Benchmark-only synthetic cohorts, one per outer fold, each generated from that fold's real training samples (TSTR, SHAP, DCR, attribute inference).
`../real_data/split_manifest.csv`	Outer-fold assignment of real samples: `sample_id`, `split`, `partition` (train or test).
`bench_metrics.yaml`	ML target, DE contrasts, anchor and quasi-identifier columns and gene counts; a reusable template for defining custom DE/ML analyses.

`artifacts/de/de_clinical_kras_mut_0_vs_1.csv`
`artifacts/de/de_clinical_egfr_mut_0_vs_1.csv`
`artifacts/de/de_clinical_stk11_mut_0_vs_1.csv`
	Per-gene DE results, one file per contrast in Table 3 order: `gene`, `gene_symbol`, real and synthetic `log2fc` and `padj`, `passes_expression_filter` (top 80% by real mean CPM), and `real_de`, `synthetic_de_strict`, `synthetic_de_relaxed` flags as counted in Table 3.

`artifacts/metrics.csv`	Summary metrics. Per-secret, per-record and per-class results are in `attribute_inference.csv`, `dcr.csv` and `clinical_fidelity_metrics.csv` in the same folder.
`../real_data/`	Real `expression.csv.gz` (samples × genes) and `metadata.csv` (clinical variables by sample ID), where the source permits redistribution.

- **Layout.** Each synthetic CSV has one row per synthetic sample. The first, unnamed column is a row number, not a real sample ID: synthetic samples do not correspond to individual real patients. The remaining columns are the clinical variables (`clinical_os_months`, `clinical_os_status`, `clinical_age`, `clinical_sex`, `clinical_egfr_mut`, `clinical_kras_mut`, `clinical_alk_mut`, `clinical_tp53_mut`, `clinical_stk11_mut`, `clinical_stage_clean`) and 19,719 gene columns.
- **Counts.** Gene values are raw, unnormalized, non-negative integer counts: no library-size normalization or log transform has been applied. Normalize them as any RNA-seq count matrix (e.g. DESeq2 size factors or log CPM).
- **Gene IDs.** Gene columns keep the source study's identifiers: 16,933 Ensembl gene IDs (e.g. `ENSG00000259429`, without version suffixes) and 2,786 gene symbols (e.g. `TMSL3` and `PKM2`), as distributed by the source; some may be retired HGNC symbols. The `gene_symbol` column of the DE files maps Ensembl IDs with a packaged HGNC table; unmapped names are kept as they are.

Limitations and intended use

This report supports a dataset-specific assessment of dbTwin under the analyses implemented above. Several limitations define its proper interpretation.

- Successful benchmarking does not imply fidelity for analyses that were not tested. In particular, this pipeline does not evaluate pathway enrichment, survival models, network inference, rare-variant or tail behavior, or replication in an external cohort.
- DE results are limited to the specified contrasts and the 15,775 most highly expressed genes. ML results use one classifier family and a target-focused feature-selection scheme. Other estimands, preprocessing choices, and model classes may behave differently.
- The five outer folds are deterministic and stratified by the primary ML target. Minority classes may still produce unstable fold-specific estimates. SHAP figures display one representative split; DE, UMAP, clinical fidelity and co-expression use the full cohorts once.
- UMAP and MMD quantify agreement after a particular nonlinear projection; co-expression is restricted to 500 variable genes. These summaries do not fully characterize the high-dimensional joint distribution.

- DCR is a memorization test, not a privacy proof. Attribute inference is an empirical, threat-model-specific attack using the stated auxiliary variables and evaluator. Neither analysis provides differential privacy or guarantees protection against linkage, genotype reconstruction, or attacks with different external information.

The synthetic cohort is therefore best suited to exploratory analysis, hypothesis generation, workflow development, and preliminary feature prioritization, followed by confirmation on governed real data. It should not be treated as a substitute for real-cohort validation in clinical, regulatory, or other high-stakes settings.

References

[1] Nanda A, Saha S. rna-twin: synthetic bulk RNA-seq cohorts and benchmark (2026). https://github.com/Nanda-Aditya/rna-twin

[2] Pilgram L, Dankar FK, Drechsler J, Elliot M, Domingo-Ferrer J, Francis P, Kantarcioglu M, Kong L, Malin B, Muralidhar K, Myles P, Prasser F, Raisaro JL, Yan C, El Emam K. A consensus privacy metrics framework for synthetic data.

[3] Giomi M, Boenisch F, Wehmeyer C, Tasnádi B. A unified framework for quantifying privacy risk in synthetic data. *Proceedings on Privacy Enhancing Technologies* 2023(2). https://petsymposium.org/popets/2023/popets-2023-0055.php

Appendix A. Per-gene statistics and library size

These standard RNA-seq checks compare the full real cohort with the full synthetic release, so, like the other full-cohort comparisons, they are in-sample.

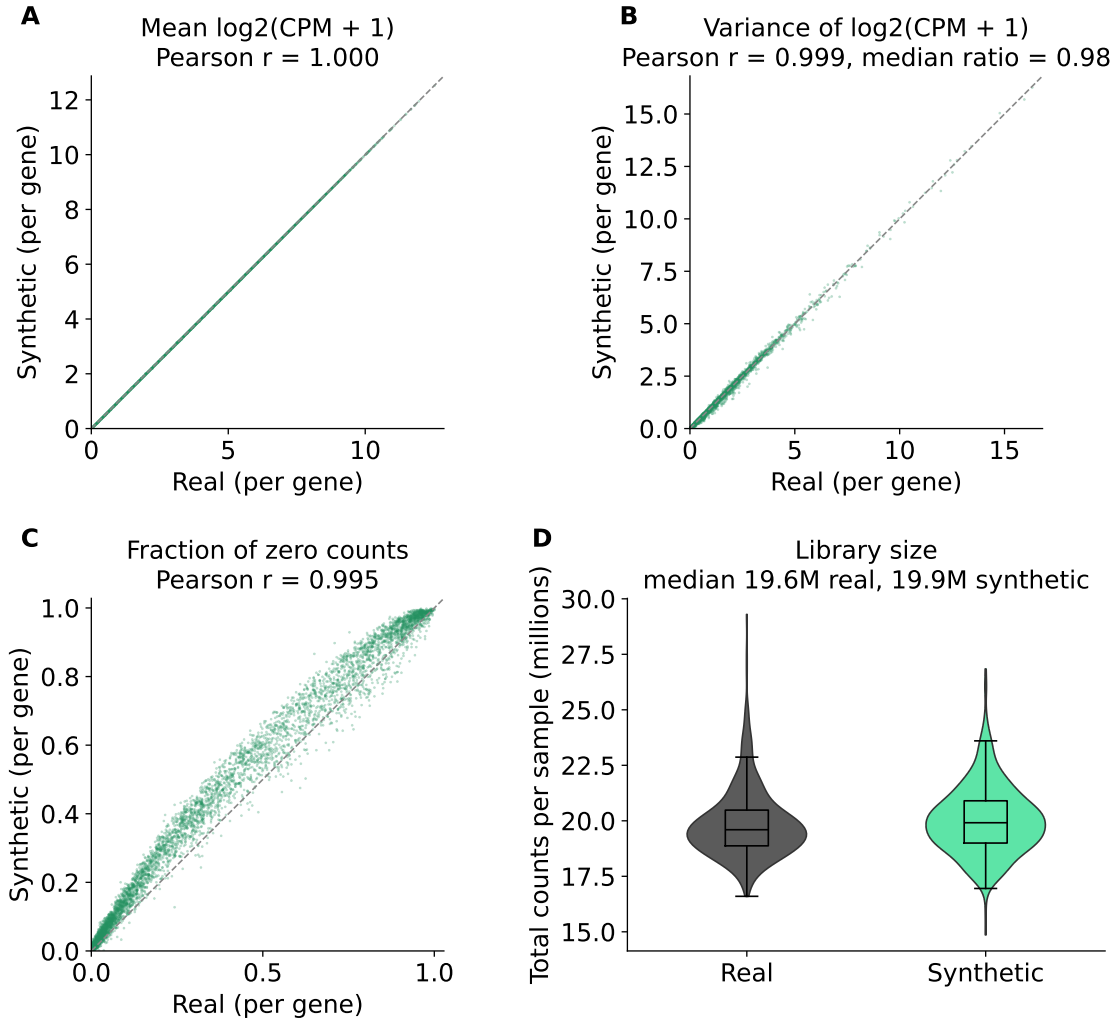


Figure A1: **Per-gene statistics and library size, full real cohort versus full synthetic release.** Per-gene statistics use log2 counts per million plus one (CPM from each sample's total counts) for all 19719 genes. (A) Mean of each gene, real (x) versus synthetic (y). (B) Variance of each gene across samples; the title also gives the median synthetic-to-real variance ratio over genes with non-zero real variance (1 = variance kept, below 1 = smoothed). (C) Fraction of samples with a zero count. Dashed lines, identity; titles give the Pearson r over genes. (D) Library size (total counts per sample) as violins with box plots (median, IQR, whiskers to 1.5 IQR).